

CprE 3810 Lab 2

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1. Prelab

Read this lab manual. In addition, read Free Range VHDL Chapters 5, 6, and 7: pages 51–68, 71–86. Describe the provided dffg.vhd in terms of edge sensitivity and reset type (active low/high and synchronous/asynchronous).

In the process statement, we can see i_CLK and i_RST are the sensitivity list items, so they will be the edges where changes are implemented.

In lines 53 and 55, we see if (i_RST = '1') and elsif (rising_edge(i_CLK)) then.....s_Q <= s_D. This tells us that this DFF's written output is synchronous with the rising edge of clock, and the reset is active high asynchronous.